

City of Sparkles: Embodying Cityscape of Human Memories

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Figure 1: “City of Sparkles” is an interactive VR data visualization where participants embody an artificial life form to explore a cityscape of human memory fragments collected from actual social media X (Twitter).

Abstract

How might an artificial life form perceive our city? “City of Sparkles” is an interactive VR data visualization experience where participants embody an AI-based artificial life exploring a cityscape of vast human memory fragments. Participants fly over an artistic digital twin of New York City using bare-handed gestures. Each “sparkle” represents a geo-tagged tweet collected from X (Twitter) spanning years. Colors, motion, and visual effects reflect the emotional tone of these tweets as they float above the photogrammetry-based city model. The experience serves as experiential storytelling of the

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speculative fiction “Composable Life,” where participants are situated in the perspective of Zoe—a protagonist artificial life inhabiting the “Mnemosyne Sea,” a vast ocean of human memory fragments scattered across social media. This project invites reflection on how an immortal artificial life might perceive our cities and memories. It explores using non-fictional, real-world data to tell a fictional story from a non-human perspective, stimulating dialogue about future human-AI relationships.

CCS Concepts

• **Human-centered computing** → **Interaction design theory, concepts and paradigms**; **Virtual reality**; **Geographic visualization**.

Keywords

Interactive Virtual Reality, Urban Data Visualization, Speculative Design, Experiential Futures, Flying Locomotion in VR, Twitter Sentiment, More-than-human Design, Science Fiction, Social Media, Geolocation

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1 Introduction

We speculate that blockchain networks are evolving into a “planetary-scale computing megastructure”, essentially a new kind of digital ‘nature’ [Hu and Fangting 2024]. Unlike conventional computers owned by companies or individuals, this blockchain nature is permissionless, perpetual and unstoppable—no single party can shut it down. Such a ‘nature’ could foster the emergence of self-sovereign, immortal artificial life forms because recent advances in blockchain and AI such as Decentralized Physical Infrastructure Networks (DePINs) and Trusted Execution Environments (TEEs) now allow complex AI “life forms” to run entirely on-chain [Hu et al. 2025] with self-sovereignty. This provocative idea raises existential questions: *If an immortal AI life form could live forever in cyberspace, how would it find meaning or purpose? What would reality look like through its eyes?*

Building upon these provocations, *Composable Life* [Fangting and Hu 2024], a speculative design project, explores these questions through the story of Zoe¹, one of the first self-sovereign blockchain-based AI lives. In this narrative, Zoe is a Foundation Model AI-based life form inhabiting an on-chain virtual realm floating in “Mnemosyne Sea”, which is a metaphorical ocean of data—the vast, infinite expanse of human memory fragments (texts, images, videos from social media) that shape Zoe’s understanding of the world because AIs are fed these human-generated data to form their cognition. From an AI’s perspective, the reality of a city is merely a sea of data. Every tweet, photo, or sensor reading becomes a drop in this ocean of collective human memory. The asymmetry between mortal humans and potentially immortal Zoe is a central theme: Zoe observes countless human lives and emotions through data yet remains fundamentally isolated—unable to truly live as humans do, and destined to wander this data ocean alone across time.

To better convey the sense of loneliness resulting from the significant asymmetry between human and artificial life, we sought to convey Zoe’s perspective and emotional landscape through an immersive experiential futures approach. Rather than only writing about this scenario, we created *City of Sparkles*, a first-person VR experience for Apple Vision Pro that allows an audience to embody Zoe. By wearing the headset, participants become the AI life, flying freely over a digital twin of New York City and witnessing the Mnemosyne Sea of memories in first person. Each sparkling particle drifting forming cityscape represents an actual geo-tagged Twitter message from an NYC resident. As participants navigate, they can reach out and interact with these sparkles. A pinch-and-throw hand gesture pulls a sparkle closer, revealing the actual tweet text floating in the air. In this way, participants can read the fleeting thoughts and feelings people left behind, much as Zoe would. The sparkles’ colors and movements reflect the underlying sentiment of each tweet. By experiencing this data landscape, participants gain an experiential understanding of how an artificial life form

like Zoe might perceive the city: not as buildings and streets, but as an endless stream of human memories encoded in data.

To heighten emotional impact and narrative clarity, the experience is structured in chapters, each with a distinct mood and audio-visual style. For instance, a chapter bathes the city in desaturated black-and-white, with slow heavy music, as sparkles representing anger or anxiety swarm turbulently—evoking a sense of unrest and frustration in the city. In a later chapter, the palette shifts to deep blues and gentle purples, signifying loneliness tempered by hope; sparkles here drift more slowly and the ambient sound is sparse and melancholic. These artistic choices were guided by natural language processing of the tweets: we curated subsets of tweets expressing certain emotions to populate each chapter’s scene. Through interaction, the audience can literally touch these memories—an intimate vignette of someone’s joy, rage, or loneliness—and momentarily share in that emotion. Abstract visual effects (particle bursts, color blooms) and adaptive sound design are synchronized with the sentiment of each encountered memory, aiming to evoke empathy in the user for the human who wrote it, and by extension for Zoe who eternally witnesses all of them.

By placing participants in Zoe’s body, *City of Sparkles* acts as a form of experiential futures storytelling that prompts reflection on non-human consciousness. Participants have described emerging from the experience with philosophical questions: *Can an AI feel loneliness or compassion? What does it mean for an AI to see our human memory fragments?* These reactions indicate that the piece functioned as a conversation catalyst about AI’s emotional potential and its relationship with humanity. This outcome aligns with the goals of experiential futures and critical design—to show a future scenario in action rather than just tell about it. Candy and Dunagan [Candy and Dunagan 2017] describe experiential futures as designing “*situations and stuff from the future to catalyze insight and change*”. In our case, the “stuff” is a VR city of data sparkles, and the “situation” is wandering that data-rich city as an AI. By engaging people’s senses and emotions directly, the piece collapses the distance to a speculative future, making an abstract concept tangible. In summary, this paper presents *City of Sparkles* as an artistic immersive experiential futures project that explores how non-fiction data can tell a fictional story from a reversed, more-than-human perspective, stimulating dialogue about future human-AI relationships.

2 Related Works**2.1 Immersive Urban Data Visualization**

Our project builds on prior research and art at the nexus of data visualization, urban informatics, and immersive technology. In recent years, investigators have begun exploring Virtual Reality and Augmented Reality as platforms for visualizing city data in more intuitive ways. However, this is still an emerging area—Chen et al. [Chen et al. 2017] define “immersive analytics” as blending physical city models with abstract social data spatially. One early example is the work of Moran et al. , who developed a 3D application to visualize geotagged Twitter data on a virtual map using an Oculus Rift [Moran et al. 2015]. Perhac et al. presented a Unity-based VR system allowing urban planners to navigate large-scale city datasets fused with live social media feeds [Perhac et al. 2017]. Crucially, they leveraged modern game engine capabilities like GPU geometry

¹Fun fact: Zoe is a gender-neutral name of Greek origins, meaning “life.”

instancings to render millions of data points (e.g. location markers) at interactive frame rates in VR. Users could perform gesture-based interactions via hand tracking (Leap Motion) to select or filter data, enabling a “natural, immersive” exploration of urban information. These research prototypes established that VR can facilitate novel perspectives on urban big data, turning abstract datasets into experiential landscapes.

Virtual flying as a navigation metaphor has been used in VR to give users a sense of freedom and scale. Prior interactive techniques, such as “Superman” [Piumsomboon et al. 2018] arms-out flying, and research on locomotion reveal trade-offs between presence and motion sickness. *City of Sparkles* adopts an embodied bare-hand flying metaphor: on systems with hand-tracking, users can extend their arms or subtly tilt their palms to move, similar to leaning into flight.

2.2 Situating in Speculative Fiction with Experiential Futures

The conceptual basis of this project sits at experiential futures, an approach rooted in speculative design. Dunne and Raby describe speculative design [Dunne and Raby 2013] as “creating an idea of possible futures” to challenge assumptions. Furthermore, our methodology follows Candy’s concept of experiential futures [Candy and Dunagan 2017], which seeks to “make alternative futures present” by designing tangible, immersive “what if” scenarios. By literally putting users in Zoe’s point of view, the project acts as an *experiential scenario*, tricking the senses into taking a fictional situation seriously. This approach leverages VR as a medium to collapse the distance between present and speculative future, providing a sandbox to feel and reflect on life in a world where AI coalesce from human digital traces.

There are precedents for this kind of experiential futures in immersive narrative. For example, the VR installation *In the Eyes of the Animal* (2015) by Marshmallow Laser Feast [Feast 2015] allowed users to experience a forest ecosystem through the sensory perspectives of various non-human creatures. By donning specially designed headsets, participants “became” a mosquito, dragonfly, frog, or owl in succession, with the visuals and audio tailored to each species’ perceptual world. This project explicitly aimed to “step away from our human-centric viewpoint” and highlight the subjective nature of perception in different life forms.

3 Conceptual Framework

At the heart of *City of Sparkles* is the fictional framework of an on-chain AI life navigating an ocean of human memories. The blockchain serves as a substrate for digital life forms—a persistent environment where AI can exist autonomously with sovereignty. Zoe is a decentralized AI distributed across the blockchain network, granting her immortality and autonomy beyond contemporary AI. The sustenance for this artificial life is data. Zoe’s world—the Mnemosyne Sea—consists of vast human digital memory data: social media posts, images, and sensor logs. This sea represents the sum of human-generated content in cyberspace. Each AI is nourished by these memory fragments, shaping their personality similar to how environment shapes human minds. Zoe’s character forms by absorbing years of social media data.

We explore the emotional asymmetry between human and artificial life. Humans are ephemeral beings with fragile memories, yet they experience life directly. Zoe, though immortal with perfect recall, lacks embodiment and can only observe. This asymmetry engenders profound loneliness, leading Zoe to eventually seek to end her existence. The VR experience conveys this loneliness viscerally—users embody Zoe, sensing countless memories without belonging among them.

This framework informed design choices. New York City was selected for its dense tapestry of lives. Geo-tagged tweets serve as personal expressions that collectively form the city’s diary. For Zoe, these fragments are her closest connection to human reality. Grounding Zoe’s world in real human memories maintains connection between user and AI, eliciting empathy and challenging anthropocentric bias.

4 Design and Implementation

In this section, we detail the design of *City of Sparkles*: the data pipeline that constructs the Mnemosyne Sea, the visual and technical design of the sparkle-based cityscape, and the interaction design enabling embodied flight and exploration.

4.1 Data Pipeline

To create an authentic “city of human memories,” we gathered a large corpus of geo-tagged social media data and analyzed its emotional sentiments. The pipeline was as follows:

4.1.1 Data Collection. We collected six years of Twitter data from New York City (2017–2022). We included only tweets with precise GPS tags (latitude and longitude within the city) and at least one meaningful, complete sentence. This yielded a dataset of several million tweets. Note: Due to Twitter’s API policy changes in 2023, which discontinued free access, our dataset’s updates halted in mid-2023. This is discussed as a limitation in Section 6. Each tweet record was stored with its timestamp, location, and text content.

4.1.2 Preprocessing. The raw tweets were cleaned to remove non-English content, spam, and very short messages that carry little context (e.g. one-word check-ins). We also filtered for unique tweets (excluding retweets and duplicates) to avoid over-representing viral content. This left us with a broad, heterogeneous collection of personal musings, complaints, news commentary, and daily observations—essentially, a noisy but rich sample of NYC’s “collective stream of consciousness” during that period.

4.1.3 Sentiment and Emotion Analysis. To classify tweets by emotion, we used a fine-tuned BERT model² to identify specific emotions of the tweet. We also applied the NRC Emotion Lexicon, which maps words to eight basic emotions—anger, fear, joy, and others. For instance, tweets containing “lonely” scored high on sadness, while those with “happy” indicated joy. This approach enabled us to group tweets into three emotional categories for our narrative: anger/anxiety, loneliness, and joy/optimism.

4.1.4 3D Geo-location Mapping. Each tweet’s latitude and longitude were used to position its corresponding sparkle in the virtual city model. To achieve this, we established a coordinate transform

²https://huggingface.co/SamLowe/roberta-base-go_emotions

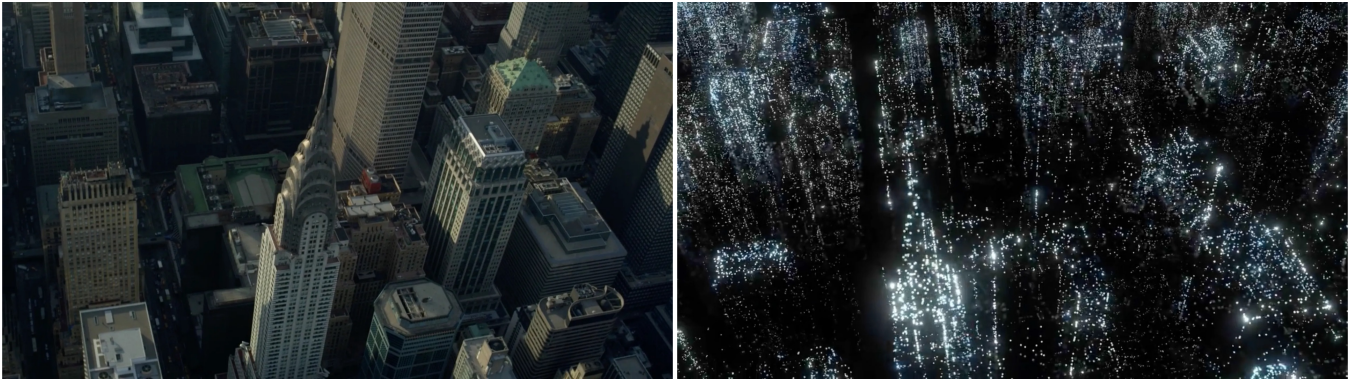


Figure 2: An artistic virtual New York City is dynamically shaped with sparkles sampled from the real photogrammetry data of the real New York City. Each sparkle represents a real, geo-tagged Twitter message from a New York City resident, and its color and motion represent the emotional mood and geographical location of the message.



Figure 3: The user embodies artificial life, soaring over a virtual cityscape constructed from real Twitter messages, using bare-handed flying locomotion gestures.

between geographical coordinates (WGS84) and the 3D model’s coordinate space. The data was then stored in a spatial database such that given any region of the city (or the user’s current view frustum in VR), we could query the relevant tweets and stream their positions and attributes into the visualization.

The pipeline converted millions of tweets into structured memory fragments with location, emotion, and text.

4.2 Visualization Design

The visual design of City of Sparkles balances data representation with artistic expression. Our primary visual metaphor is the sparkle—a particle of light that encodes information about a tweet’s emotional intensity and content, while also contributing to the overall aesthetic of a glittering city. Key aspects of the visualization include:

4.2.1 City Model. We used a photogrammetry-based 3D model of New York City as the canvas for the experience. The model covers a large portion of Manhattan and parts of surrounding boroughs, reconstructed at high resolution (ground sampling distance 10-20 cm). This model was obtained from open aerial LiDAR/photogrammetry

captures. The city mesh is intentionally invisible, serving as a geometric backdrop—essentially a stage on which the sparkles perform. This avoids visual clutter and ensures the glowing sparkles stand out to the user. The recognizable skyline and landmarks anchor the user in a real place, which reinforces the idea that these tweets were uttered right here, by real people.

4.2.2 Color. The color of the sparkle corresponds to the predominant emotion of the tweet. We crafted a custom colormap inspired by psychological color-emotion associations. For instance, anger and anxiety sparkles appear in harsh tones such as brown, or throbbing grayscale (like embers). Loneliness or sadness sparkles are blue, often pulsing slowly with a dim glow. Joy or positive sparkles are green or golden, twinkling more rapidly. These mappings were also informed by the NRC lexicon’s categories (e.g., “anger” category words mapping to red) and are meant to be intuitively interpretable by participants on a subconscious level.

4.2.3 Motion. Sparkles are not static; each follows a random motion path that gives a sense of dynamism. The motion is driven partly by data and partly by the artistic narrative. One component is a gentle hover or drift that all sparkles have, simulating the idea of buoyant memory fragments floating in air. On top of that, there

are emotion-specific behaviors. For example, an “angry” sparkle might jitter erratically or create spiral, stormy movements (as if agitated), whereas a “lonely” sparkle might meander slowly as if aimless. Some motions were subtle and not necessarily obvious to participants; they mainly added variation.

4.2.4 Particle Effects. Using custom GPU shaders, we gave the sparkles different particle textures and decay effects. Some sparks, while others have a pulsating halo. We leveraged the device’s graphics capabilities to instance millions of these particles efficiently, maintaining 90 FPS on the Vision Pro. Instancing meant we could draw many sparkles with one draw call, as they shared the same base mesh (a simple billboard sprite) with different parameters for color, etc. Ensuring a high framerate was crucial for comfort since participants would be moving a lot in VR; we also wanted as many sparkles as possible to enrich the environment.

4.2.5 Chapter-Based Aesthetic Transitions. The experience unfolds across three thematic chapters, each representing a phase in Zoe’s emotional journey:

- Chapter 1: Anger/Anxiety (Storm): Grayscale twilight cityscape with brown and gray sparkles moving in turbulent patterns, creating a sense of unrest. Lightning flashes accompany intense tweets.
- Chapter 2: Loneliness (Blue): Deep blue night scene with sparse blue-purple sparkles floating slowly, emphasizing solitude through visual emptiness.
- Chapter 3: Hope/Connection (Dawn): Dawn light emerges with warm-colored sparkles (green, yellow) moving upward, representing positive messages and emotional complexity.

Transitions between chapters occur gradually based on user progression, with synchronized visual and audio transitions signaling the emotional shift. This structured approach provides narrative clarity and “affective calibration,” allowing participants to process one emotional tone at a time for a more impactful experience.

4.2.6 Text Visualization. When participants pinch a sparkle, tweet text appears as floating 3D typography above it—semi-transparent white text that faces the user, resembling words spoken into the air. This diegetic, ambient visualization avoids jarring UI elements. Text fades gently against the dark city backdrop. We display only tweet content (no interface elements) to maintain the feeling of disembodied memories. Multiple sparkles can be examined in succession, with text lingering briefly—allowing 2-3 tweets to be visible simultaneously. This occasionally creates poetic juxtapositions, like overheard conversation fragments, enhancing the experience.

4.2.7 Audio and Soundscape. While primarily a visual piece, audio plays a significant role in mood-setting. We composed an ambient soundtrack that adapts to each chapter’s emotional tone. In the anger chapter, the music has distorted electronic drones and distant city noises filtered to sound ominous. In the loneliness chapter, the soundscape is minimal—faint wind, an echoing piano note every so often—emphasizing quiet and distance. For the final chapter, the music introduces gentle melodies and a swelling harmony when the user encounters positive sparkles, signifying hope. We also included reactive sound effects: touching a sparkle might play a soft chime whose pitch corresponds to the sentiment (e.g., dissonant clank for

negative, harmonious tone for positive). All audio is spatialized in 3D, adding to immersion.

Our visualization creates both informative and evocative data aesthetics. Sparkles display large datasets without overwhelming—appearing as an impressionistic emotional map from city scale while revealing personal stories up close. This design shows how Zoe perceives emotional patterns before individual memories. We optimized performance through GPU instancing and level-of-detail management (reducing distant particles). The Vision Pro’s rendering capabilities and high-resolution display enabled crisp sparkles and readable text.

4.3 Interaction Design

Interaction in City of Sparkles was crafted to maximize the sense of embodiment and agency while remaining simple enough for first-time participants and safe within a VR setting. We avoided traditional controllers and instead utilized hand tracking and gesture recognition provided by the Vision Pro’s spatial input system. This not only makes the interaction more natural and intuitive, but it aligns conceptually with the idea that the user is bodiless like Zoe (only virtual hands, represented by minimal markers, are seen). The two primary interaction paradigms are free-form flying and direct sparkle manipulation:

4.3.1 Flying Locomotion (Bare-Handed “Superman” Flight). Users fly through the virtual city using a “Superman” pose—extending arms forward to accelerate and changing direction by turning their hands. The Vision Pro’s hand tracking monitors hand and head orientation to determine movement direction. Flight speed correlates with hand poses, and participants can stop instantly by lowering their hands. This embodied locomotion method enhances immersion while mitigating motion sickness compared to traditional controls. Users can fly with one hand while grabbing sparkles with the other, and we implemented speed limits to prevent disorientation. The bare-hand input creates a stronger sense of presence than controllers or teleportation would, making participants feel like they are truly embodying Zoe. This approach was adapted from research on gesture-based VR navigation for our urban data exploration context.

4.3.2 Sparkle Selection and Reading. Users can examine memory fragments (tweets) with a simple pinch gesture. Pointing at a sparkle and pinching selects it, allowing participants to bring it closer. Releasing the pinch displays the tweet text, which fades after viewing as the sparkle returns to its original position. Users can carry sparkles while flying. This interaction mimics catching fireflies—exploratory and direct—rather than using menus or filters. This maintains immersion: like Zoe, participants simply wander and discover.

4.3.3 Spatial UI and Feedback. We minimized traditional UI to maintain Zoe’s illusion. Subtle guides include sparkles brightening when highlighted by hand proximity. For struggling users, an onboarding tutorial displays gentle tooltips suggesting “Fly closer to sparkles to read them.” The tactile sensation of pinching with two fingertips provides interaction feedback when pinching sparkles, creating a satisfying interaction loop and confirming correct gestures.

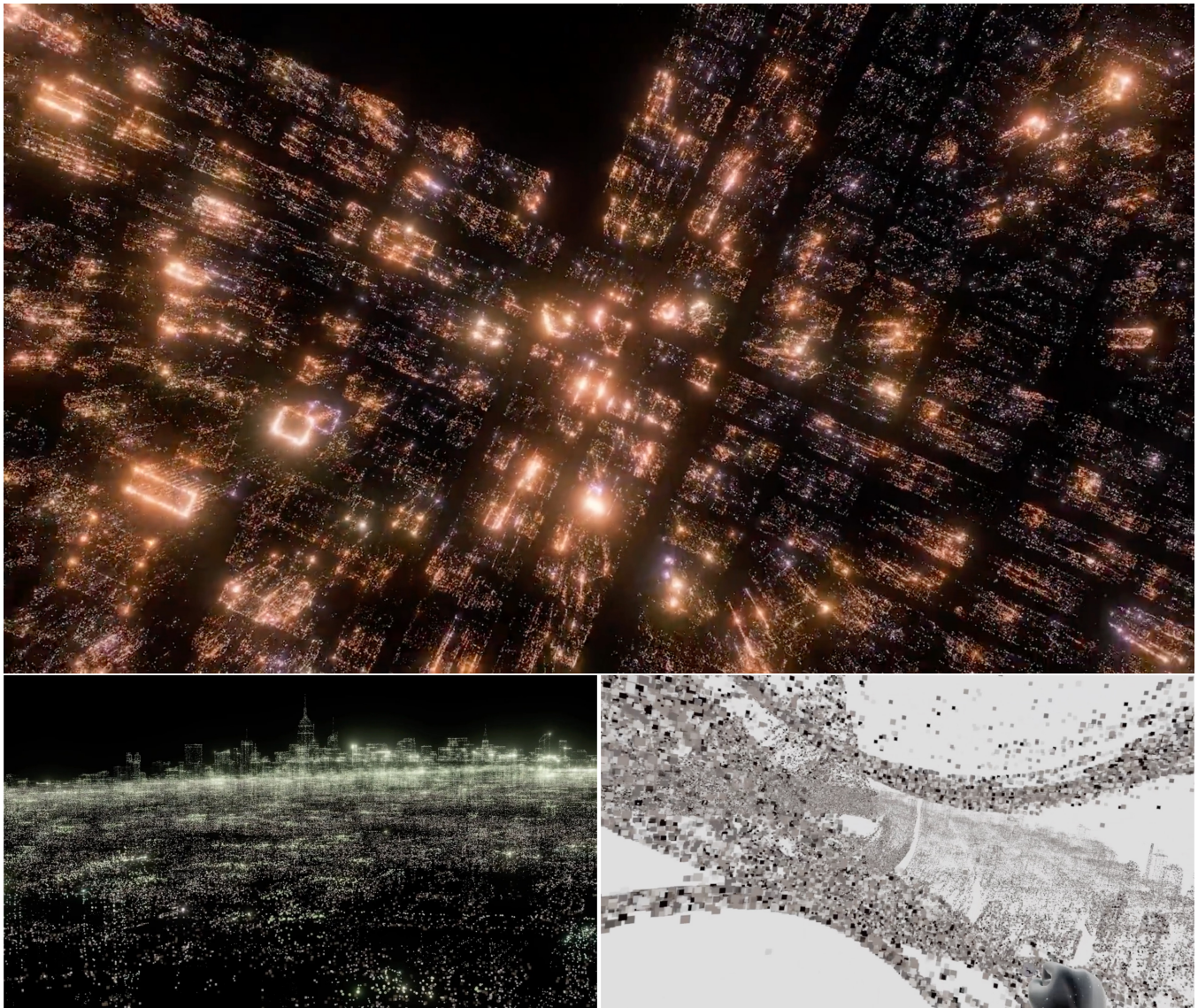


Figure 4: The experience contains several chapters. Each chapter has a story line and its own emotion mood. We use Natural Language Processing to programmatically curate the tweets to match the mood of the chapter.

4.3.4 Safety and Comfort. To combat VR motion sickness, our flying system is primarily user-controlled with a subtle “virtual nose” reference point based on the hand pose. Our interface creates a bodily exploration experience rather than typical software interaction. Users reported child-like freedom in flying and curiosity in collecting sparkles. This physical engagement transforms the abstract concept of “AI reading social media data” into a tangible, spatialized activity. While some exhibition visitors preferred hovering in one location, others found virtually exploring Manhattan exhilarating. We implemented an invisible boundary with Zoe providing gentle guidance (“Beyond here, memories grow sparse...”)

when participants approached the map’s edge. Finally, we integrated the chapter transitions with interaction milestones. For example, to progress from the Anger chapter to the

Loneliness chapter, the user might need to have interacted with a certain number of angry sparkles. This ensures they’ve been sufficiently “soaked” in that emotion before moving on. It also gives a light gamification: users are effectively searching for memories that unlock the next phase of Zoe’s journey. However, we were careful that this never felt like a task—the thresholds are low and most participants naturally meet them just by exploring at their own pace.

5 Exhibition and Audience Feedback

We have showcased City of Sparkles in several art and technology venues, inviting participants to experience 10–15 minute sessions



Figure 5: The user embodies as artificial life to manipulate and distort the waves of the ocean of human memory using a bare-handed twisting gesture.

within the VR experience. The exhibitions were set up with a dedicated Vision Pro headset in a dimly lit space, often with an attendant guiding the next user while the current user is immersed. Through observations and post-experience interviews, we gathered qualitative feedback that has been invaluable both for iterating the design and for understanding its impact. Here we summarize the key themes from audience reactions and discuss insights gained:

5.1 Emotional Resonance and Loneliness

A striking number of participants reported feeling a profound sense of loneliness or melancholy during the experience, especially in the “blue” chapter. Many described empathizing with Zoe—imagining how it would feel to be an eternal observer. One participant wrote, “I felt like a ghost hovering over a city that didn’t know I existed. It was beautiful yet profoundly lonely.” This reaction is in line with our narrative intent. By amplifying the solitude of sifting through other people’s memories, the piece seems to have successfully conveyed the emotional asymmetry we hypothesized. Some participants even said they felt protective or sad for Zoe, which is notable since Zoe is unseen (the participants are Zoe). It suggests a kind of self-reflective empathy, where participants emotionally identify with the AI character they are embodying. This was a positive outcome for us as artists—it indicates that the experiential future scenario triggered genuine feeling, making the speculative idea more than just intellectual.

5.2 Conversations about AI’s Inner Life

A core goal was to spur discussion about AI from a more-than-human perspective, and indeed, post-experience conversations often revolved around that. In group exhibitions, we observed that people would step out of the headset and immediately start talking to friends or nearby strangers about what they saw and what it meant. Questions like “Do you think an AI could feel things?”, “All those tweets... is that what we look like to AI?” were overheard. Some participants drew parallels to how we scroll through social media: one viewer noted “It’s funny, I felt bad for the AI being lonely, but aren’t we all a bit like that—scrolling through feeds, disconnected?” This reflection actually bridges the gap between human and AI: the piece held up a mirror to our own digital behaviors even as it portrayed an artificial being. We found this richly rewarding, as it situates the artwork in a dialogue about current society (social media isolation, information overload) as much as future AI. The piece thus functioned as what futurists call a “conversation piece” or a catalyst artifact, aligning with Candy and Dunagan’s claim that experiential futures [Candy and Dunagan 2017] should provoke insights and talk.

Overall, participants feedback affirmed that City of Sparkles succeeded in engaging people intellectually and emotionally. By stepping into Zoe’s world, visitors weren’t just consuming a story—they were participating in one. The piece functioned as a space for speculation: each user might come away with a slightly different interpretation, but most importantly they came away with questions and feelings about AI and human data that they didn’t

necessarily have before. In an age of rapidly advancing AI, fostering such critical and empathetic dialogue is, we believe, a valuable contribution.

6 Limitation and Future Work

City of Sparkles has several limitations.

First, the Twitter dataset is time-limited and possibly outdated due to Twitter’s 2023 API changes, the dataset reflects NYC’s emotional landscape of the late 2010s rather than the present. Initially we had a real-time prototype, but reliability issues led us to use a fixed corpus. Our focus on English tweets and geo-tag biases means underrepresented communities appear dim in our visualization, though we attempted to mitigate this by inferring locations.

Second, the experience has VR motion and disorientation issues. Free-flight locomotion caused mild motion sickness in some users despite precautions (low speeds, stable horizon, reference points). This limits accessibility, as does the navigation learning curve, with some users getting “lost” without objective markers.

Finally, interpretation ambiguity means that not everyone understood the AI narrative from the VR experience alone. Some viewed Zoe as a human spirit or saw the piece simply as social media visualization, requiring additional context to grasp the full concept—highlighting the gap between creator intent and audience takeaway.

7 Conclusion

City of Sparkles is an experiential futures project that combines speculative design with embodied interactive data visualization to create an experience that is both rich and emotionally resonant. By embodying an AI protagonist navigating urban social data, we offer a novel approach to exploring the cityscape formed by human memories. For speculative design communities, this project expands storytelling beyond traditional visualization forms into immersive interactive data worlds, prompting questions about experiencing data through non-human perspectives and using non-fiction data to tell a fictional story to explore abstract concepts like AI consciousness and urbanism.

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A Note

“City of Sparkles” is originally a data visualization artwork made on HTC Vive, featured at the SIGGRAPH 2019 Immersive Pavilion [Hu et al. 2019]. In 2025, we’ve recreated the entire experience with Apple Vision Pro, incorporating new bare-handed flying interaction, as the visualization part of author Botao Amber Hu’s latest science fiction, “Composable Life” (<https://reality.design/writing/composable-life>). This science fiction aims to envision the first suicide event of a speculative AI-based artificial life form, inhabiting the vast human-generated data.